Ajay Kumar Garg Engineering College
Department of Computer Science & Engineering

PROJECT ID
CSE-2026-033
YEAR
2025 – 26

AI-Powered Personalized Mental Health Companion

Leveraging DistilBERT for Real-Time Emotion Detection and Support

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01 INTRODUCTION

With over 970 million people globally facing mental health challenges, accessing professional therapy remains difficult due to high costs, social stigma, and a severe shortage of practitioners. To bridge this gap, we developed an AI-Powered Personalized Mental Health Companion. Utilizing a fine-tuned DistilBERT natural language processing model, this system accurately detects user emotions from text in real-time. Moving beyond rigid rule-based chatbots, it acts as a digital first-responder—providing personalized, evidence-based coping strategies while ensuring strict data privacy and incorporating a critical crisis-detection safety net

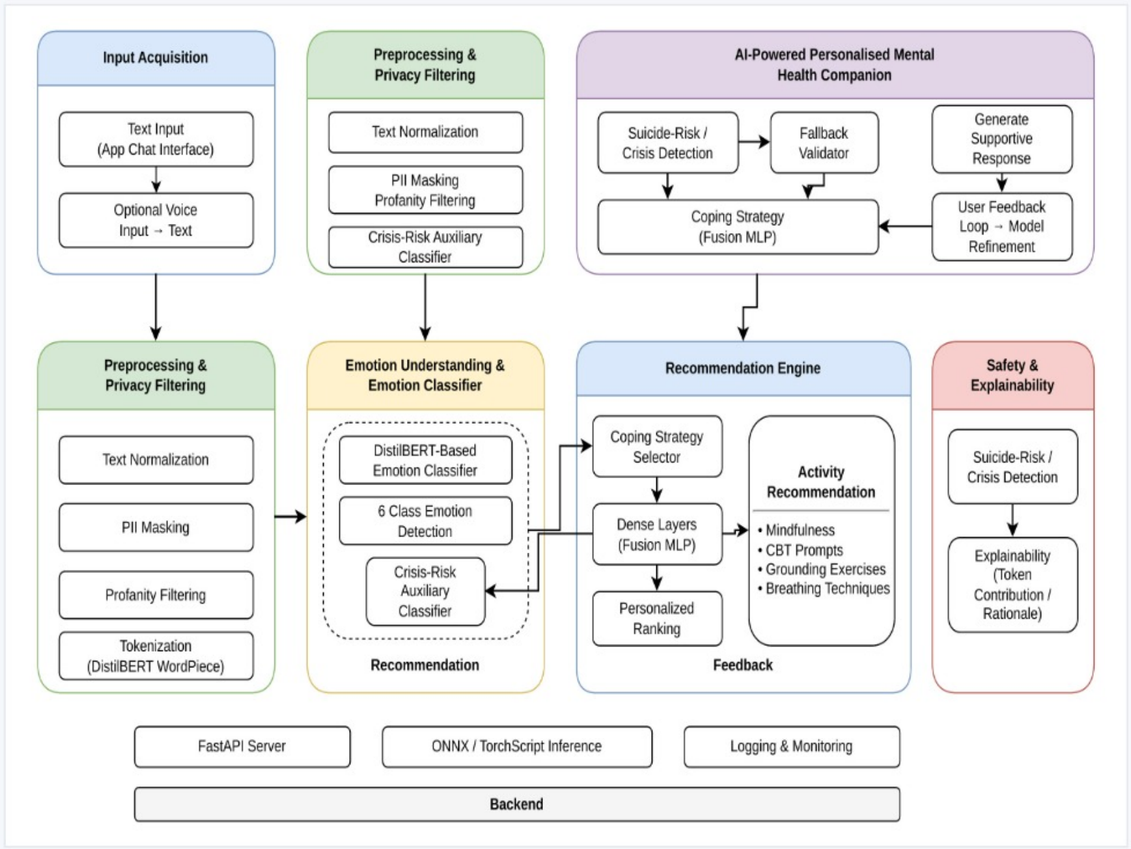
02 OBJECTIVES

- Accurate Emotion Detection: To develop a highly accurate NLP model (DistilBERT) capable of identifying 6 core human emotions from text.
- Personalized Support: To provide tailored coping strategies (like mindfulness or CBT) dynamically based on the user's current emotion and past feedback.
- User Safety First: To implement a robust "Crisis Detection" system that immediately identifies severe distress and provides emergency resources.
- Data Privacy & Ethics: To ensure a secure, zero-knowledge architecture where user data is encrypted and anonymity is maintained.

03 METHODOLOGY

1. Dataset & Preprocessing: Trained on the DAIR-AI dataset, using SMOTE to balance classes and text normalization for clean input.
2. AI Model: Fine-tuned a lightweight DistilBERT transformer for fast and accurate 6-class emotion detection.
3. Recommendation Engine: Delivers personalized coping strategies based on the detected emotion and adapts via user feedback.
4. Tech Stack: Built with Python, FastAPI (Backend), Streamlit (Frontend), and optimized with ONNX for low-latency.

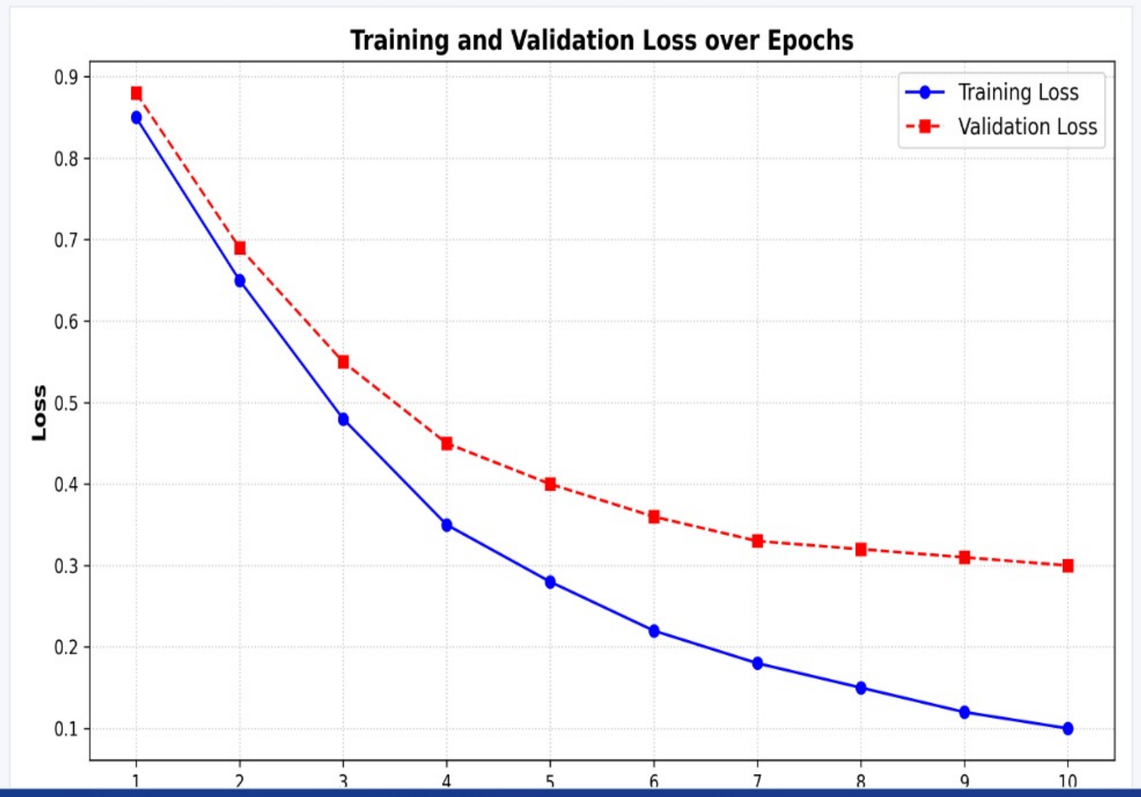
04 ARCHITECTURE



End-to-end NLP pipeline: User Input (Text/Voice) → Preprocessing (Masking/Filtering) → DistilBERT Emotion Classifier → Recommendation Engine

05 RESULTS

- Training loss drops from ~0.85 to 0.10 over 10 epochs.
- Validation loss plateaus near 0.30 by epoch 10, indicating stable learning without severe overfitting.
- Model achieves an 84% accuracy in emotion classification.



06 CONCLUSION

- Validates DistilBERT for highly accurate (84%), real-time emotion detection.
- Successfully implements a dynamic, feedback-weighted recommendation engine.
- Proves that AI mental health support can be private, safe, and accessible.

07 FUTURE WORK

- Multimodal emotion recognition (voice/facial analysis).
- Federated learning for enhanced user privacy.
- Multilingual support for broader accessibility.
- Integration with wearable health devices.

08 REFERENCES

[1] JMIR Form. Res., 2021 — Conversational AI for Mental Health Apps.

[2] NPJ Digit. Med., 2022 — NLP for Mental Illness Detection.

[3] NPJ Digit. Med., 2023 — AI Chatbots for Mental Health.

[4] ACM Trans., 2023 — Human-centered AI for Mental Health.

Scan to View Demo

NLP · DistilBERT · Mental Health · Sentiment Analysis · FastAPI